Parametric Investigation of the Operating Limits and Performance Effects of Anode Off-Gas Recirculation in Ammonia-Fueled Solid Oxide Fuel Cell Systems

Jaeyoung Park [a], Kwangnam Jeong [a], Taewook Lee [a], Jongsup Hong [a,*]

[a] *School of Mechanical Engineering, Yonsei University, 50 Yonsei-ro, Seodaemun-gu, Seoul 03722, South Korea*

[*] Corresponding author. Tel.: +82 2 2123 4465; Fax: +82 2 362 2736
Email address: jongsup.hong@yonsei.ac.kr (Jongsup Hong)

Supplementary Table S1 Performance parameters of the validated ejector model.

Ejector parameter	**Description**	**Parameter value [%]**
η_p	Isentropic efficiency regarding the choking of primary flow	99
η_s	Isentropic efficiency regarding the choking of secondary flow	23.6
φ_p	Frictional loss coefficient of primary flow	88
φ_m	Frictional loss coefficient of mixing process	93.6

Geometries of the ejector (nozzle throat diameter, constant area section diameter) are determined in each operating condition to satisfy the desired mass flow rates under critical mode operation, given the performance parameters provided above.

Supplementary Table S2 Stream data of system base case operation.

Stream number	T [°C]	p [atm]	$\dot{m}$ [kg/h]	Chemical composition [mol fraction]				
				H_2	H_2O	O_2	N_2	NH3
1	45.0	2.777	14.81	0	0	0	0	1
2	456.8	1.115	46.82	0.088235	0.41176	0	0.16667	0.33333
3	600.0	1.072	46.82	0.44092	0.3089	0	0.24994	0.00024003
4	700.0	1.067	64.01	0.13235	0.61765	0	0.25	0
5	700.0	1.067	32.00	0.13235	0.61765	0	0.25	0
6	700.0	1.067	32.00	0.13235	0.61765	0	0.25	0
7	25.0	1.000	663.12	0	0	0.21	0.79	0
8	32.2	1.074	663.12	0	0	0.21	0.79	0
9	600.0	1.074	663.12	0	0	0.21	0.79	0
10	700.0	1.068	645.91	0	0	0.19109	0.80891	0
11	700.0	1.068	415.12	0	0	0.19109	0.80891	0
12	700.0	1.068	230.81	0	0	0.19109	0.80891	0
13	457.5	1.036	230.81	0	0	0.19109	0.80891	0
14	693.0	1.036	677.92	0	0.054203	0.17341	0.77238	0
15	693.0	1.036	564.12	0	0.054203	0.17341	0.77238	0
16	693.0	1.036	113.79	0	0.054203	0.17341	0.77238	0
17	55.1	1.000	564.12	0	0.054203	0.17341	0.77238	0
18	167.7	1.000	677.92	0	0.054203	0.17341	0.77238	0

Supplementary Table S3 Stream data of ejector back pressure – entrainment stream temperature dependence analysis. (Stream properties of both flows refer to the ejector primary and entrained flow of base case operation.)

Stream Data							
Stream	***T*** **[°C]**	***p*** **[atm]**	**$\dot{m}$ [kg/h]**	**Chemical composition [mol fraction]**			
				H_2	**H_2O**	**N_2**	**NH3**
Primary flow	45	2.777	14.81	0	0	0	1
Entrained flow	Varies	1.067	32.00	0.13235	0.61765	0.25	0

Supplementary Table S4 Summary of ejector back pressure – entrainment stream temperature dependence analysis.

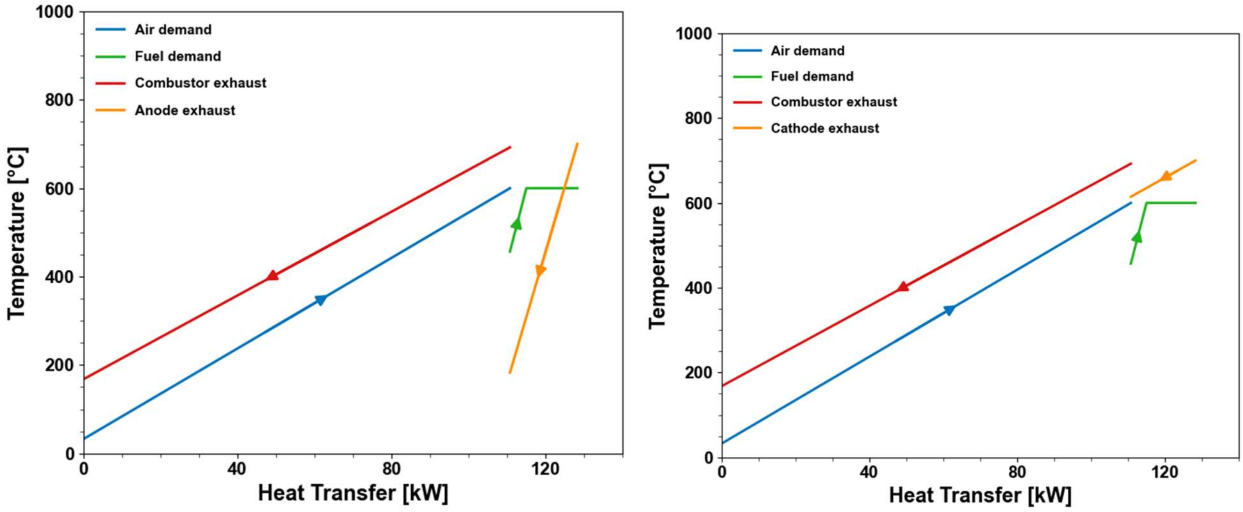
Entrained flow temperature [°C]	Mixed flow Mach number	Ejector back pressure [atm]
700	1.041	1.115
750	1.037	1.110
800	1.033	1.105
850	1.030	1.100
900	1.026	1.096

Supplementary Table S5 Summary of fuel reformer heat duty, LMTD and hot flow mass flowrate at different AOGR ratio. (System operation under base case FU_{stack}, j, T_{stack}.)

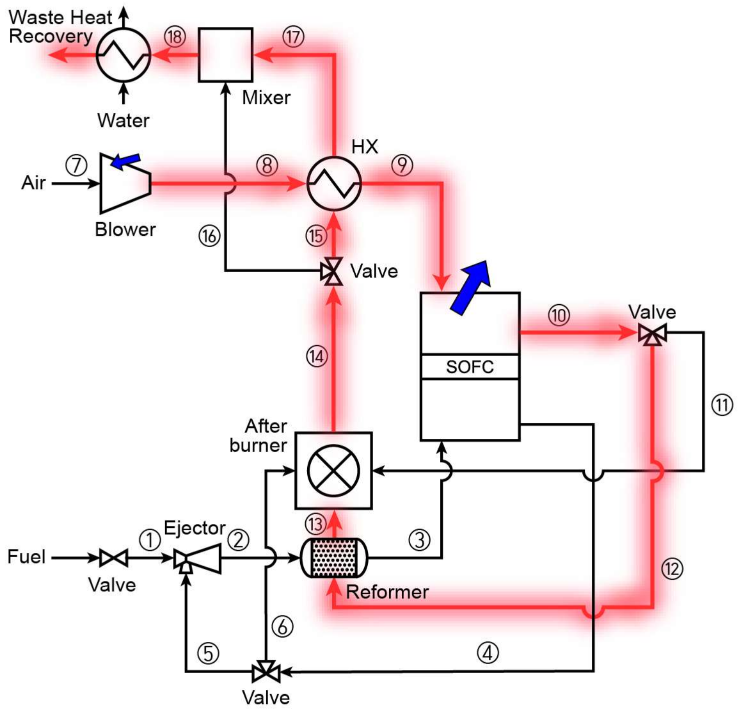
AOGR ratio [%]	Reformer heat duty [kW]	LMTD [℃]	Required mass flow rate of reformer hot flow [kg/h]
0	22.58	99.81	135.34
10	21.69	52.59	150.61
20	20.80	46.24	168.13
30	19.80	40.06	189.92
40	18.70	34.52	215.44
50	17.4	19.66	230.81
60	15.99	16.54	290.45
66.6	14.75	10.23	311.53

Supplementary Table S6 Summary of unit cell voltage at different T_{stack} operation under base case FU_{stack}, j, AOGR ratio

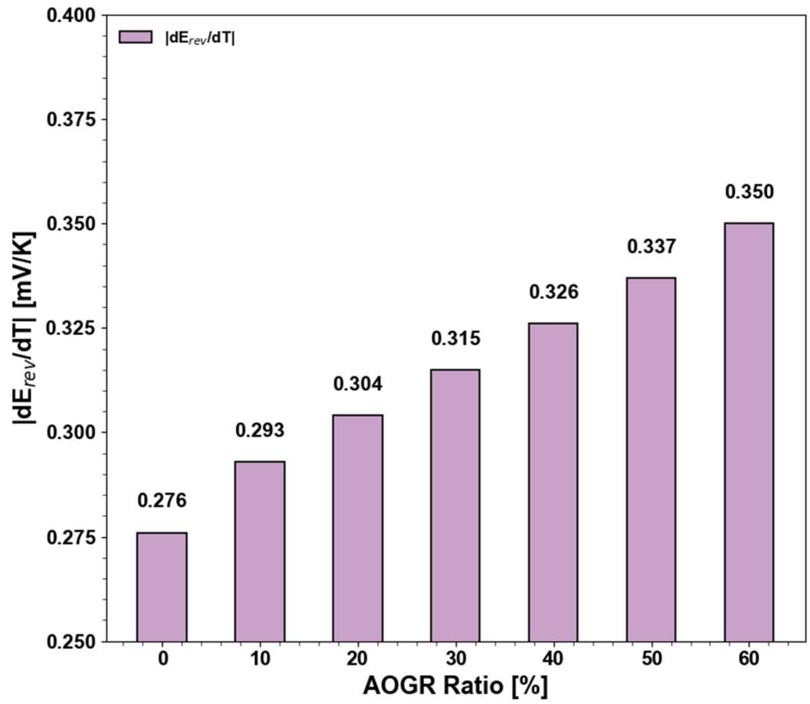
		T_{stack} [℃]				
		600	**650**	**700**	**750**	**800**
Reversible potential		0.9688	0.9521	0.9353	0.9184	0.9013
Overpotential	Activation	0.0520	0.0322	0.0197	0.0124	0.0081
	Ohmic	0.0506	0.02699	0.0152	0.0090	0.0056
	Concentration	0.0024	0.0025	0.0025	0.0025	0.0026
	Total	0.1051	0.0617	0.0375	0.0240	0.0163
Unit cell voltage		0.8637	0.8905	0.8978	0.8944	0.8850



Supplementary Fig. S1. T-Q diagrams of systems utilizing reformer heat source from: (a) anode off-gas; (b) cathode off-gas.



Supplementary Fig. S2. Flow path of fluid to be pressurized by air blower in a recirculation system.



Supplementary Fig. S3. The correlation between AOGR ratio and magnitude of E_{rev} drop sensitivity to T_{stack}.

$|dE_{rev}/dT|$ values are defined as the total change in E_{rev} drop divided by the range of whole sweep of T_{stack} (600℃~800℃).

Supplementary Note S1 Descriptions of heat exchanger sizing at different operating conditions

The shell-and-tube heat exchangers were initially sized at the base-case condition. The shell diameter was selected to maintain the cold-side fluid velocity within the recommended range of 15–30 ms^{-1}, based on conventional heat-exchanger design guidelines [1]. The exchanger length was fixed at seven times the shell diameter – following the design guidelines of conventional shell and tube heat exchangers - and the number of tubes was increased with the available tube-bundle area as the shell diameter increased [2].

Because the stream flow rates varied across the parametric conditions, the base-case geometry was resized when the calculated cold-side velocity fell outside the prescribed range. If the velocity exceeded 30ms^{-1}, the shell diameter was increased in increments of 0.05 m; if it fell below 15ms^{-1}, the shell diameter was reduced by the same increment. The exchanger length and tube number were then recalculated according to the same geometric rules, and this procedure was repeated until the velocity criterion was satisfied. The first geometry satisfying the criterion was selected. Therefore, the exchanger dimensions were not independently optimized at each operating point, but were determined using a consistent, guideline-based discrete sizing policy applied throughout the parametric study.

[1] J.E. Edwards, Design and Rating of Shell and Tube Heat Exchangers, MNL 032A, P&I Design Ltd., Teesside, UK (2008).

[2] K.Thulukkanam,Heatexchangerdesignhandbook,CRCpress,2000.